Design and Implementation of a Student Placement and Attachment System

A Case of Catholic University of Eastern Africa

- CMT 400: Project Final Defense Report
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Presentation Overview

- Background of the Research
- Problem Statement
- Main and Specific Objectives
- System Analysis
- System Design
- System Implementation
- System Demonstration

Background of the research

- Industrial attachment is a compulsory component of undergraduate training in Kenyan universities.
- enables students to apply classroom knowledge and gain practical industry experience.
- It Students are required to maintain logbooks and undergo supervision and assessment during attachment.
- The current manual process is inefficient, causing delays in placement, communication, and assessment.
- At CUEA, attachment management relies on paper-based processes and fragmented communication.

Problem Statement

- Use of traditional manual systems at CUEA has led to increased frustrations among students, supervisors, and employers.
- The system lacks efficiency in:
  - Securing attachments.
  - Manual documentation.
  - Automated supervisory evaluations.

  Solution Required: A system that streamlines and automates all activities throughout the attachment period

Main Objective

- To develop a web-based Student Placement and Attachment System

Specific Objectives

- Enable students to view and apply directly to attachment opportunities.
- Implement automated documentation of daily/weekly reports by attachées and assessments from supervisors.
- Enable real-time tracking of student progress during attachments.

System Analysis

Functional Requirements of the System

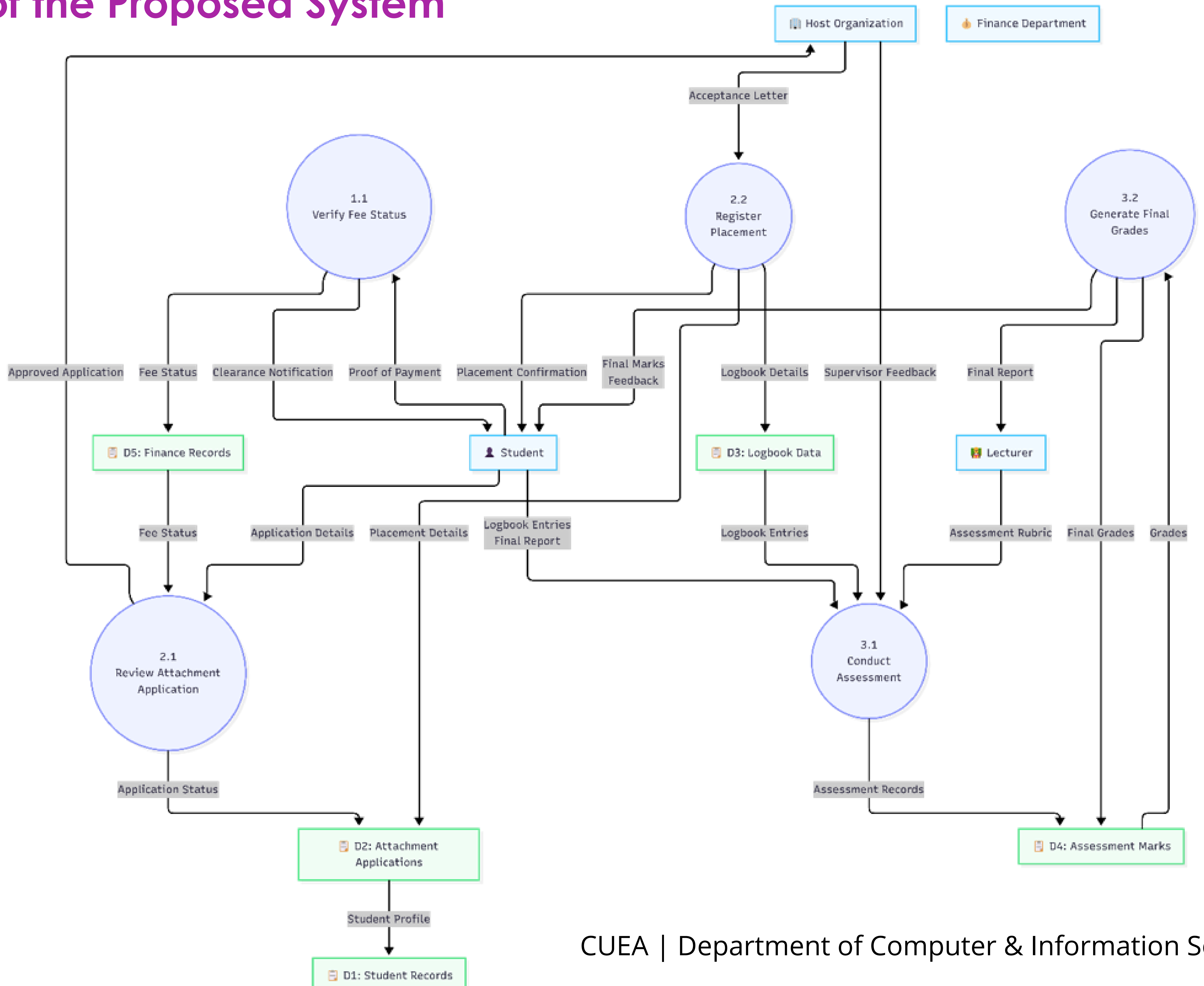
1. **Attachment Opportunity Management:** The system shall provide a centralized platform for listing companies offering attachment opportunities and enable students to view and apply directly.
2. **Student Progress Tracking**: The system shall allow students and supervisors to document and track progress throughout the attachment period, including submission of daily and weekly reports.
3. **Report Submission and Verification:** The system shall support submission, storage and verification of reports by students and supervisors.
4. **User Authentication and Access Control:** The system shall provide secure login for students, supervisors and administrators, ensuring access rights are appropriately managed.
5. **Notifications:** The system shall notify students and supervisors about application status, report submissions and important deadlines

System Analysis

Non-Functional Requirements of the System

1.**Usability:** The system shall be user-friendly and intuitive for students, supervisors, and administrators.
2. Data Security and Integrity: The system shall ensure confidentiality, integrity and protection of user and organizational data.
3. Reliability: The system shall be consistently available during operational periods.
4. Scalability: The system shall be capable of handling increasing numbers of users and attachment records.
5. Accessibility: The system shall be accessible through standard web browsers without the need for specialized software.

Level 2 DFD of the Proposed System



System Analysis

System Users in the System.

1. **Students** – visit the secretariat, submit details, find host organizations manually, maintain logbooks and submit final reports.
2. **Host Organizations / Industry Supervisors** – receive applications, provide feedback on student performance and supervise daily tasks during the industrial attachment
period.
3. **Lecturers / Assessors** – conduct physical assessments, interview students and assign marks which are then added to the final transcript.
4. **Attachment Coordinators / Secretariat Staff** – register students, issue logbooks, collect reports and coordinate assessments.
5. **Finance Department** - clear students in order to obtain logbooks

System Analysis

System Inputs in the Current System

1. **Students**

- Student Details – Name, Admission No., Phone, Faculty, Course
- Attachment Details – Organization, Start & End Dates
- Financial Details – Fee Statement
- Logbook Entries – Daily & Weekly Activities
- Final Report – Summary of Experience & Learning

2. **Host Organizations**

- Supervisor Feedback – Performance Evaluation & Comments

3. Lecturers / Assessors

- Logbooks – Reviewed Entries with Supervisor Comments
- Supervisor Feedback – Student Performance Reports

4. Attachment Coordinator

- Student Applications – Personal & Academic Details
- Organization Data – Placement Opportunities
- Application Forms – Submitted for Approval

System Analysis

System Outputs in the Current System

Students

- Application Status
- Fee Clearance
- Assessment Marks & Feedback
- Logbooks & Final Report

Secretariat / Coordinator

- Clearance Instructions
- Application Notifications
- Placement Records
- Logbook Issuance

Overall System

- Verified Attachments
- Consolidated Reports & Results

Finance Department

- Fee Statements
- Payment Clearance

Host Organizations

- Performance Feedback
- Signed Logbooks

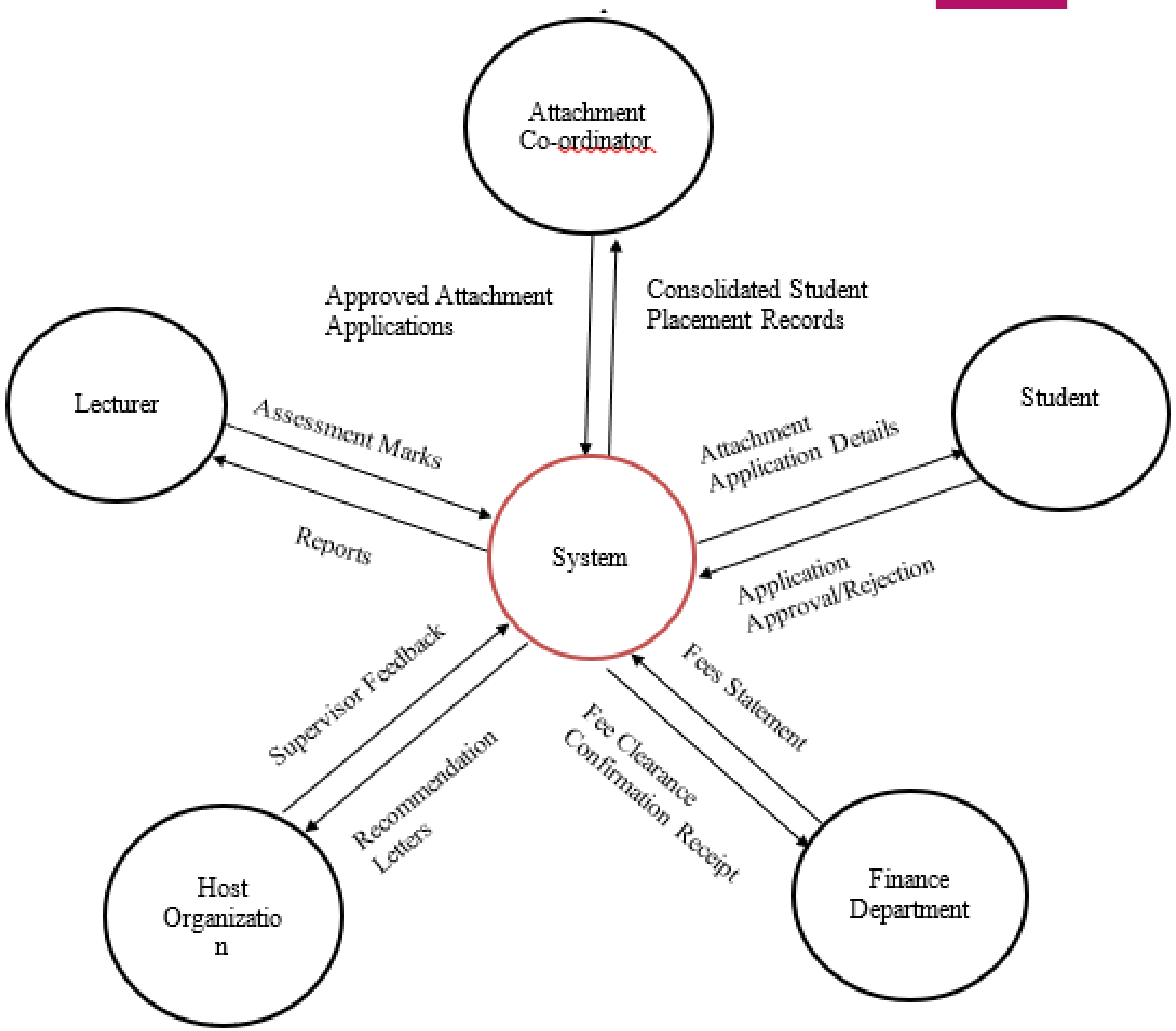
Lecturers / Assessors

- Assessment Marks
- Student Feedback

System Analysis

System Entities

- Student
- Attachment Application
- Host Organization
- Supervisor
- Lecturer / Assessor
- Attachment Placement
- Logbook
- Logbook Entry
- Final Report
- Assessment / Marks
- Faculty & Course

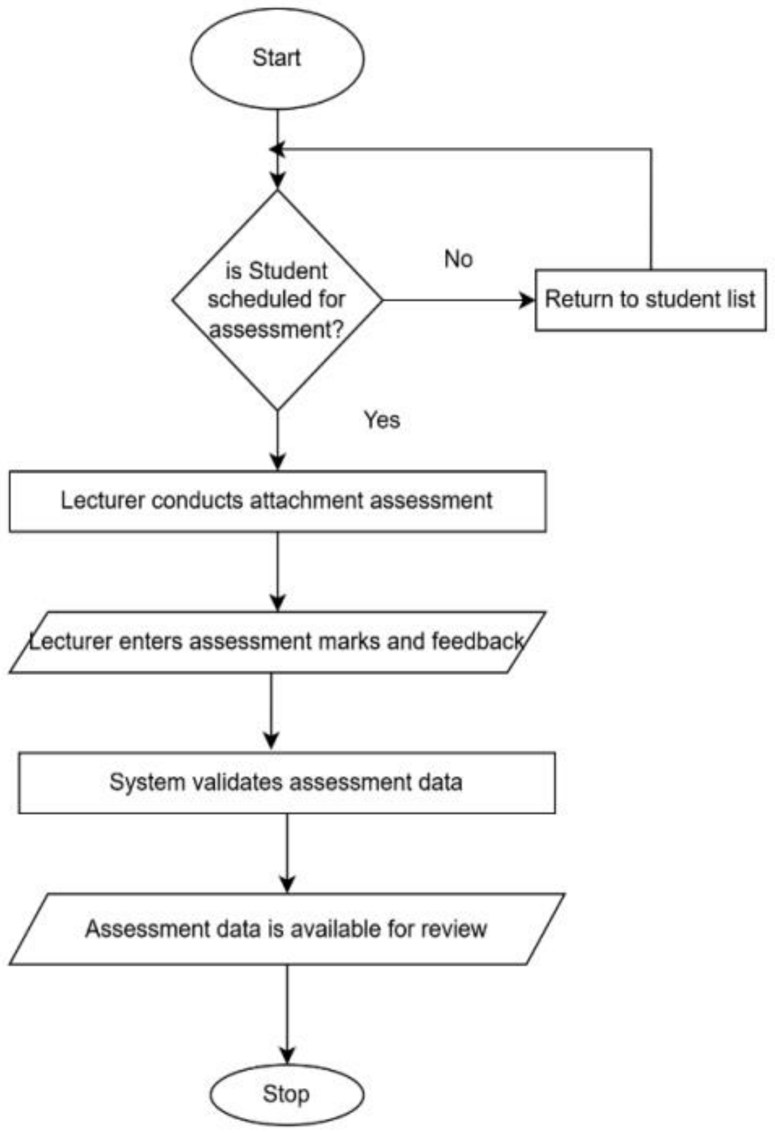


System Design

Process Design

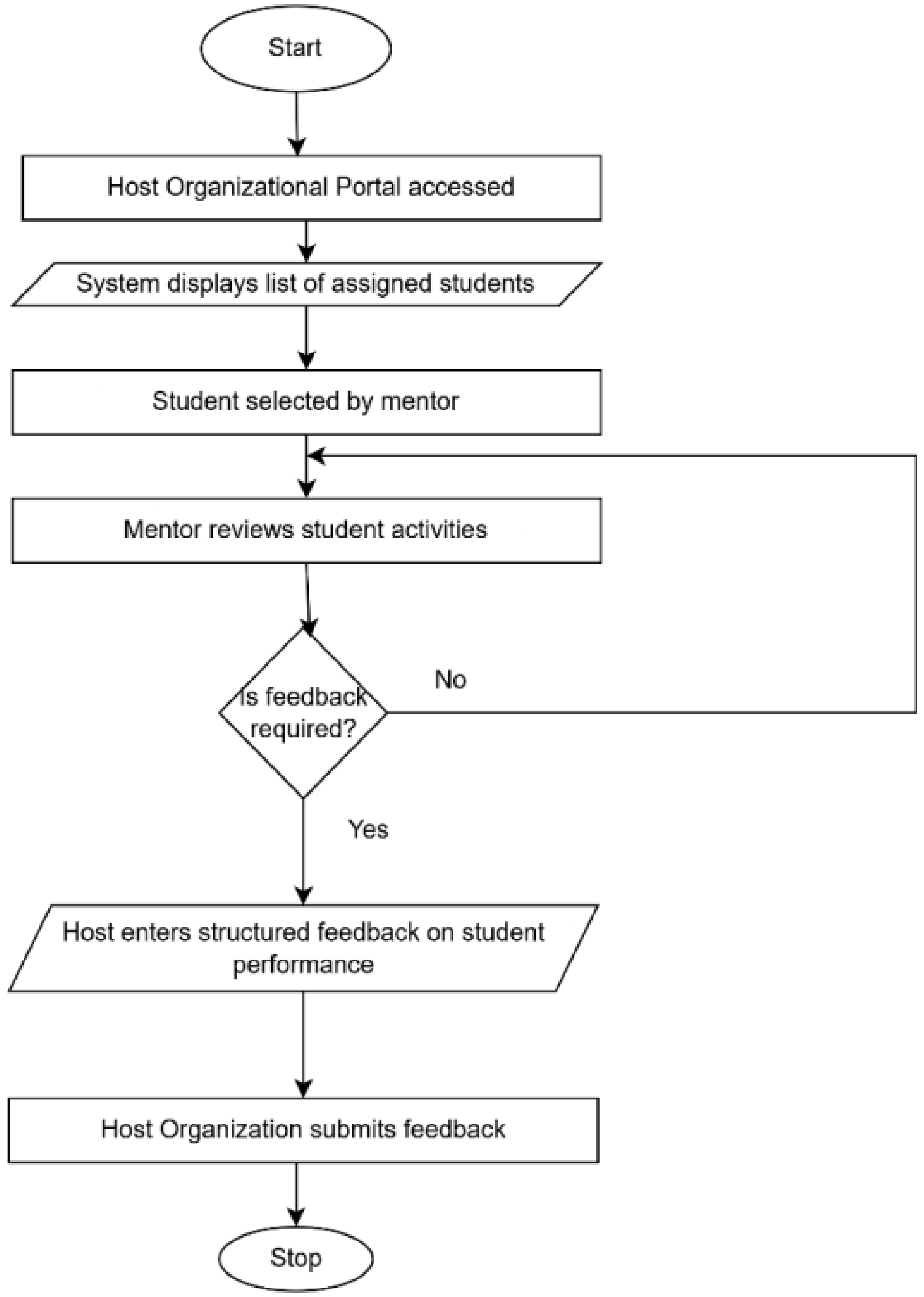
Student Assessment Workflow

This takes place once the student has been on attachment for at least a month

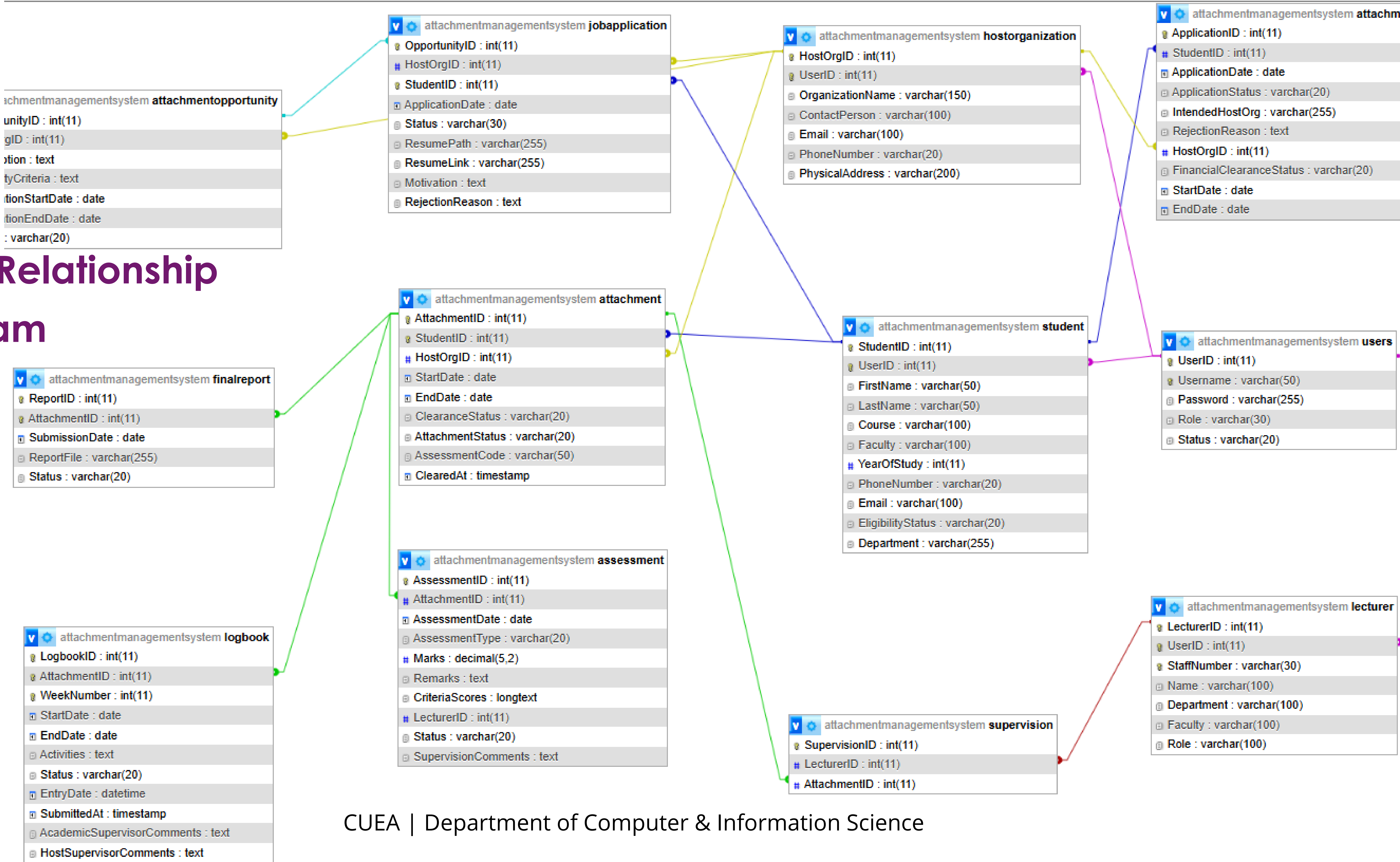


Validate Logbook Entries Flow

This flow enables the host organizations to provide structured feedback to students on attachment.



Entity Relationship Diagram



Test Data

- Simulates real-world data for system testing.
- Validates CRUD operations (Create, Read, Update, Delete).
- Tests relationships between database tables.
- Ensures data integrity and foreign key constraints are maintained.

Attachment Table Test Data

AttachmentID	StudentID	HostOrgID	StartDate	EndDate	Clearance Status	Attachment Status	Assessment Code	Cleared At
1	1	1	2026-02-01	2026-05-01	Cleared	Ongoing	ASMYO1	2026-05-02 09:30:00

System Implementation

Implementation Process

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP with MySQL as the database used

Key Features

- User authentication & role management
- Attachment application & opportunity management
- Logbook submission & supervision
- Bulk upload of students and lecturers, as well as bulk assigning of students to supervisors
- Automated student assessment & final report submission

Testing: Unit testing, Integration testing, Functional testing and User Acceptance Testing (UAT)

The End

System Demonstration and Q&A